



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Cognitive Approaches

Inform the design of systems that are intuitive, efficient, and aligned with human mental models

Information Processing Model: Equates the human mind to a computer, where information is processed in stages, including **perception, encoding, storage, and retrieval** - breaks down how users take in information from an interface, interpret it, and decide on actions

Mental Models: Internal representations of **how designers believes a users mind and system works** - built from prior experiences and interactions with similar systems

Distributed Cognition: Cognition not only as an individual process but as something that is distributed across people, artifacts, and the environment - **shared understanding of system accross many cognitions**

Embodied cognition and affordance: Thinking and understanding are not just brain-based but also rely on our body's actions, movements, and sensory experiences - Our understanding of objects is rooted in our bodily interactions, and affordances are cues that leverage this understanding, guiding our actions in ways that feel natural and intuitive

Cognitive Approaches

Inform the design of systems that are intuitive, efficient, and aligned with human mental models

Basis: Examination of the physiological and psychological aspects of awareness

Aim: “Good” software supports people in their perception and manipulation of relevant aspects of the environment

Differentiation: “How do we perceive?” vs. “How do we understand?”

? cognitive scientific approaches

Cognitive considerations are the foundation of software ergonomics / usability engineering

More examples: Attention-Memory model, Human-processor model

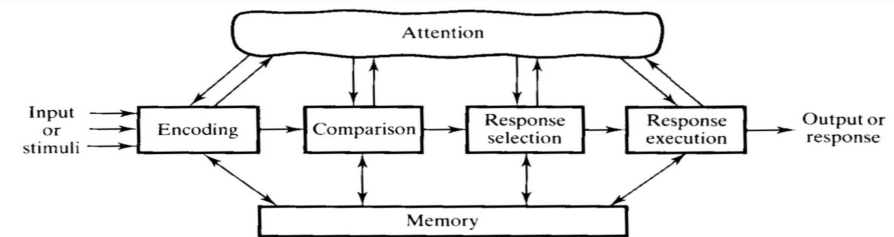


Figure 3.2 Extended stages of the information processing model (adapted from Barber, 1988).

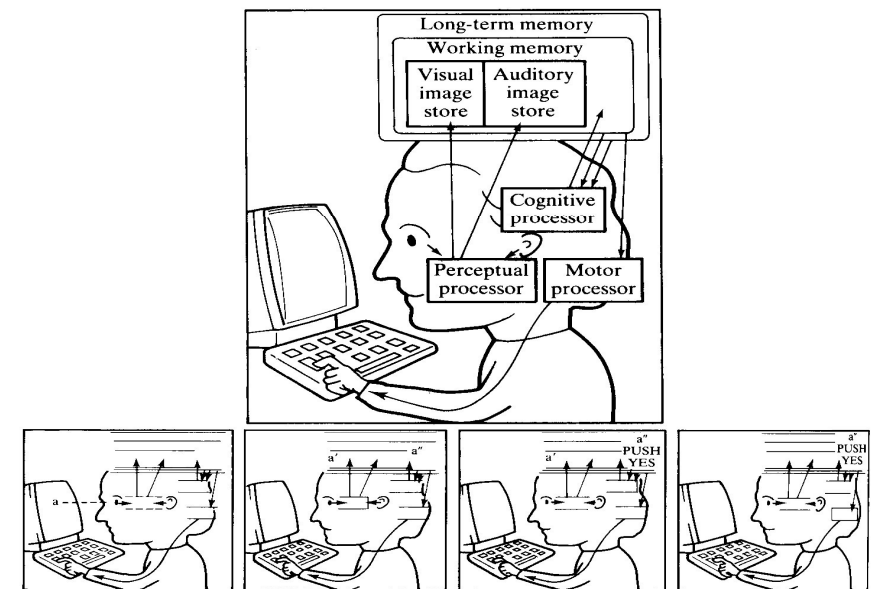


Figure 3.4 The human processor model (adapted from Card *et al.*, 1983).

Cognitive: Functional Approaches

Goal-Directed Design (Cognitive Engineering)

Focuses on understanding users' goals and the cognitive steps they take to achieve them. By identifying user goals, tasks are broken down into logical steps that align with how users think.

What is functional?

- Specific to activity, task or goals
- Practical rather than attractive

Aim:

- Models of human activities and tasks
- Improve execution
- Reduce error and frustration

Good for

- Highly structured systems
- Fixed procedures

Problem is

- Study of task execution may not be enough in some situations
- Not suitable to investigate **wicked problems**

Methods: Task Analysis

Advanced: Hierarchical Task Analysis

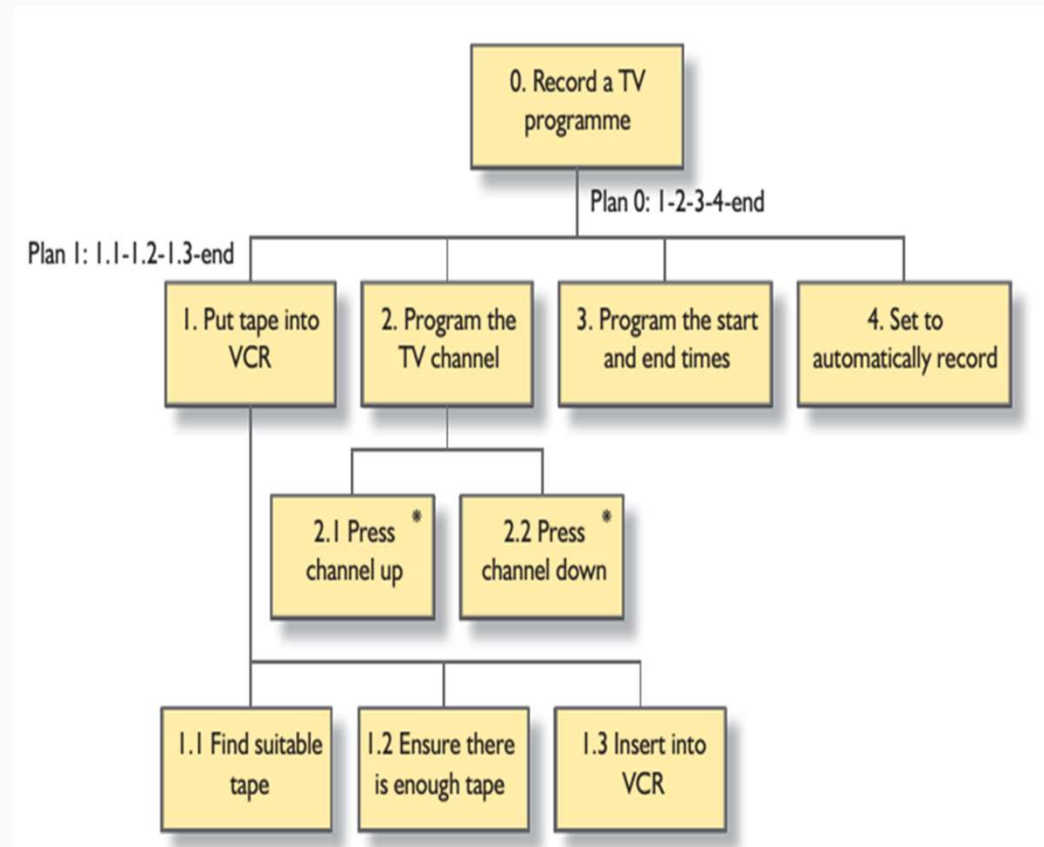
Don Norman (1998) said, users are notoriously bad at clearly articulating goals: e.g., "I want to be a good mom" – where do you even begin? Each subtask should be specified in terms of objectives. Put together, these objectives should cover the whole area of interest—i.e., help a user achieve a goal in full.

Goals: State to be achieved

Tasks: Goal + Actions

Actions: Simple performance steps

Once you have laid all the steps out, you will then be in a position to see where additional user support is required (for example, you might wish to automate some actions that the user currently undertakes), or eliminate unnecessary steps, in order to minimize the number of actions that a user has to undertake, unassisted.



Methods: Task Analysis

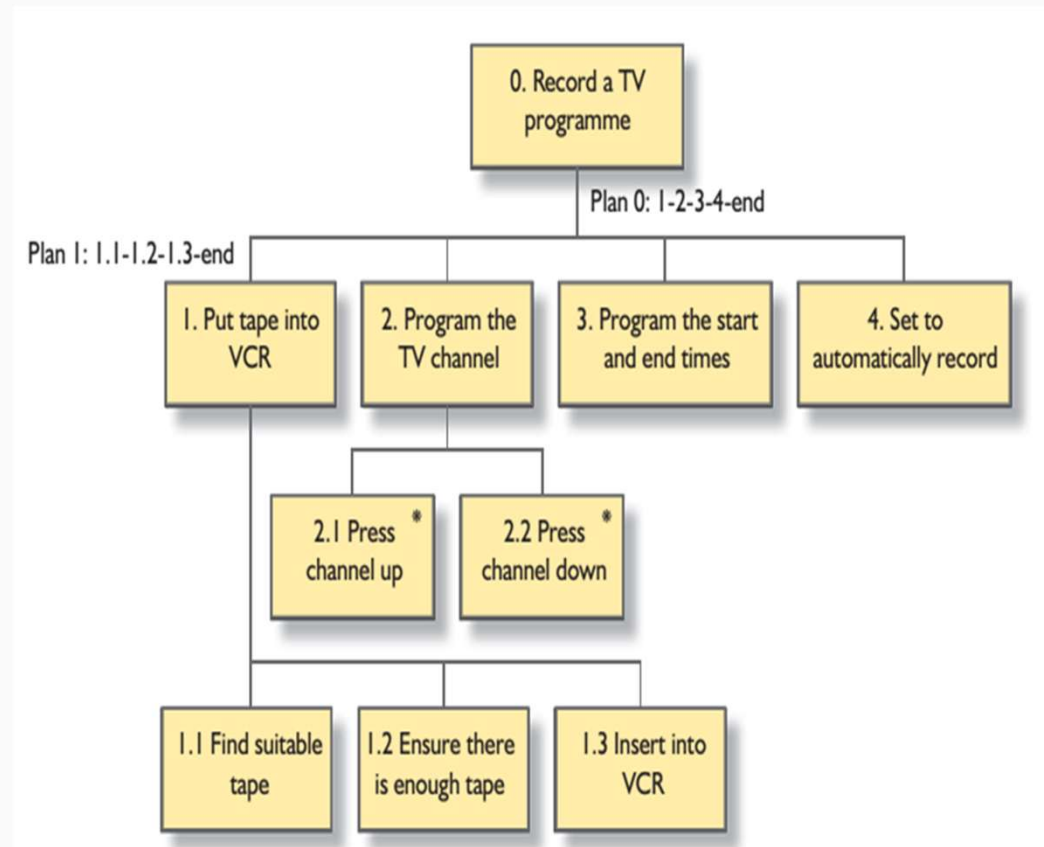
Advanced: Hierarchical Task Analysis

Goals: State to be achieved

Tasks: Goal + Actions

Actions: Simple performance steps

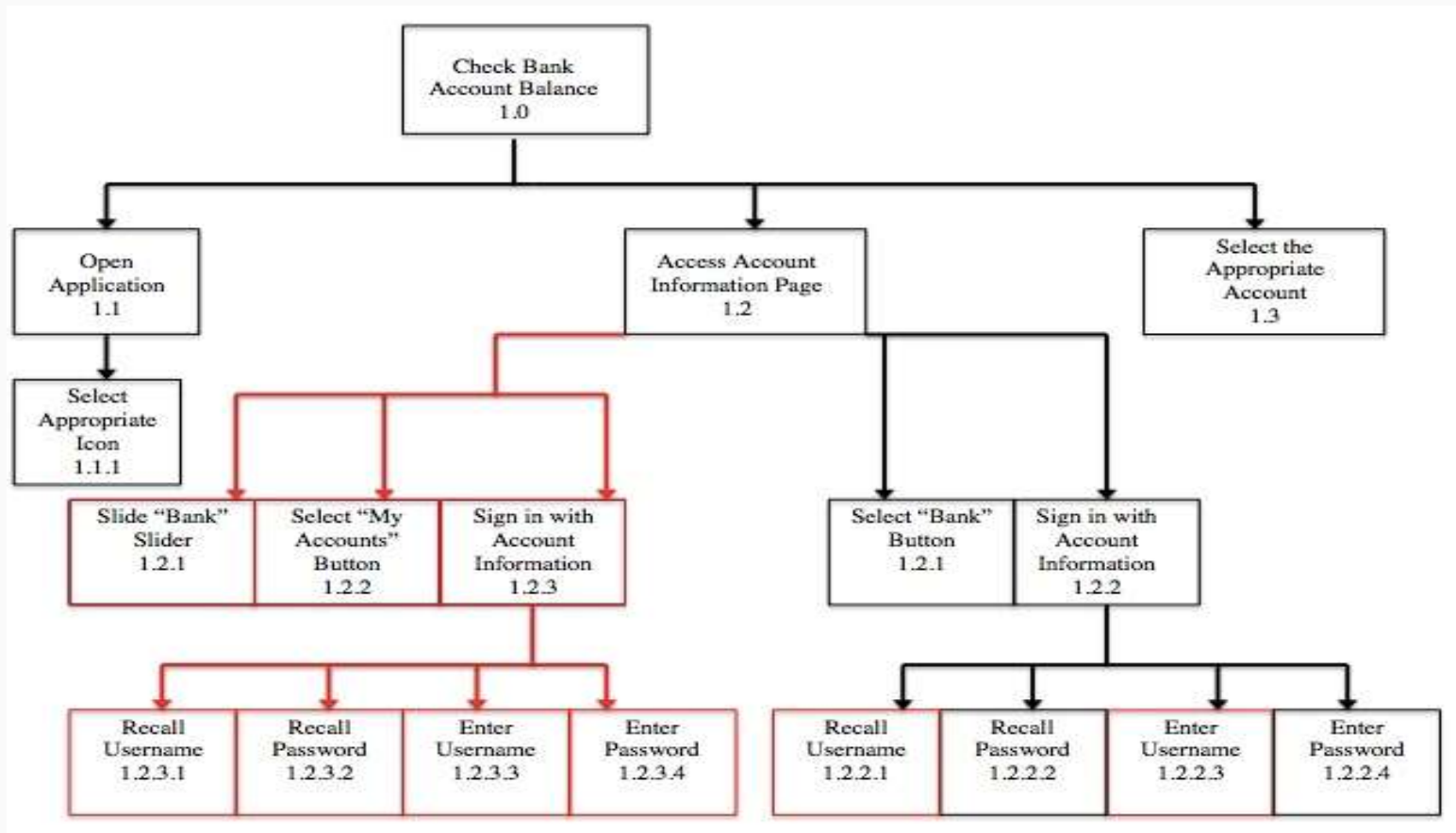
- It is read from top to bottom.
- The rectangles depict the various tasks with the top one representing the overall task.
- From this point, the tree branches out into higher-order subtasks, which, in turn, branch out into sub-subtasks, becoming increasingly granular until the level of “interactions” is reached (e.g., “Insert into VCR”).
- So-called Plans specify the sequence in which tasks on the same hierarchical level are carried out.
- Tasks may be executed in order based on their sequence and numbering, in any order, or only if certain conditions are met.



(Benyon 2010)

Methods: Task Analysis

Advanced: Hierarchical Task Analysis

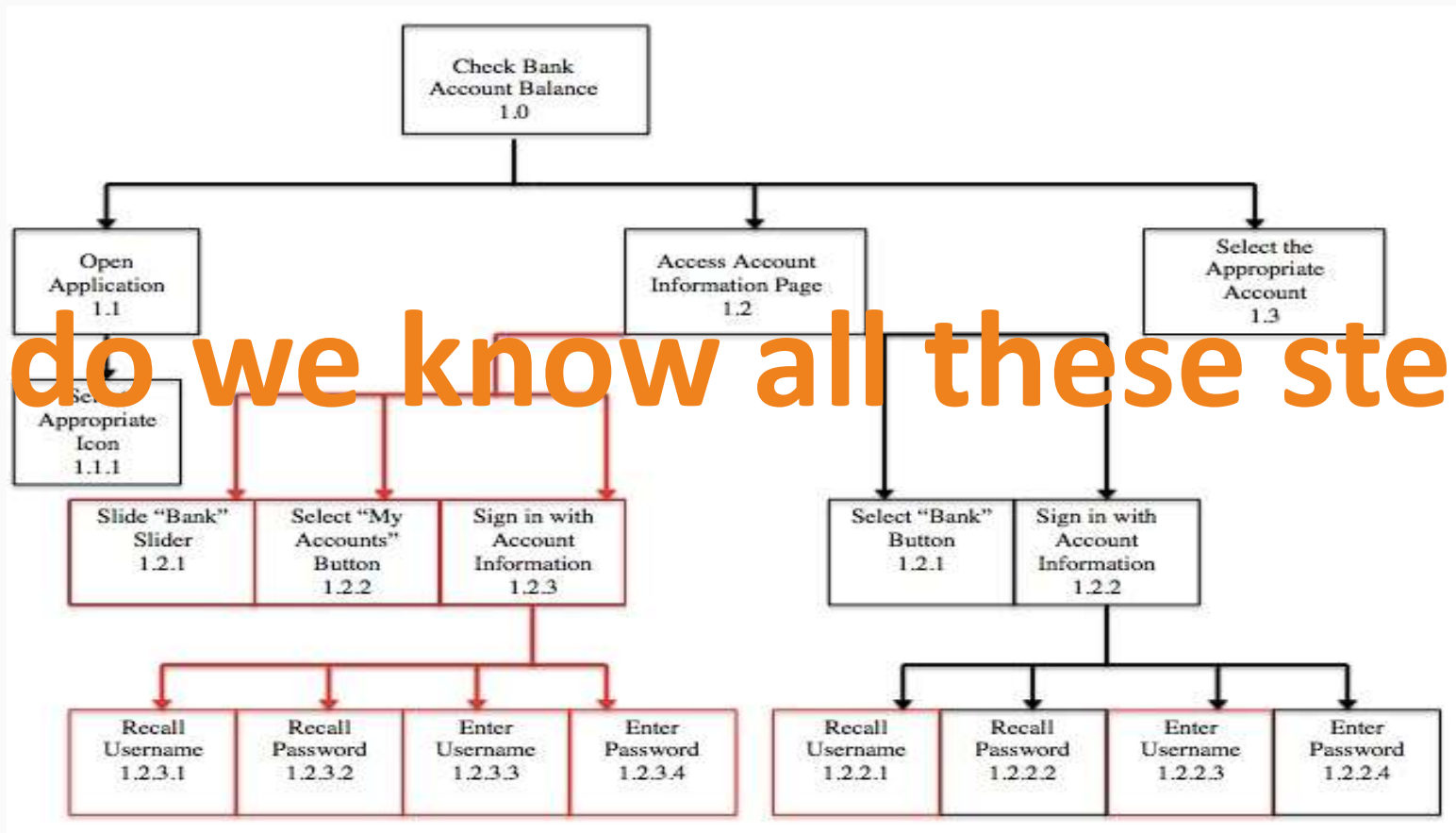


(<https://hfacmethods.wordpress.com/hierarchical-task-analysis/>)

Methods: Task Analysis

Advanced: Hierarchical Task Analysis

How do we know all these steps?



(<https://hfacmethods.wordpress.com/hierarchical-task-analysis/>)

Methods: Task Analysis

Advanced: Hierarchical Task Analysis

Goal: Send an Email

1.1 Compose Message

- 1.1.1 Open Email Client
- 1.1.2 Create New Message
- 1.1.3 Address Recipient
- 1.1.4 Compose Message Content
- 1.1.5 Attach Files (if necessary)
- 1.1.6 Proofread Message

1.2 Send Message

- 1.2.1 Review Message
- 1.2.2 Click Send Button
- 1.2.3 Wait for Confirmation

1.3 Manage Sent Items

- 1.3.1 Save a Copy in Sent Items
- 1.3.2 Close Email Client

Methods: Task Analysis

Advanced: Hierarchical Task Analysis

Main Problems

- Hard to accommodate tasks completed by more than one individual, except in very simple cases.
- Representation of a task analysis is complex, even when a simple task is studied.
- Complex tasks (e.g. air traffic control) become very unwieldy very fast, but can be done, but their interpretation is often limited to those that conducted the analysis.

Hierarchical Task Analysis	Cognitive Task Analysis	Logical task analysis
• Involves decomposing tasks into subtasks and representing the order and structure through structure diagrams. • Performing HTA is a time-consuming process with many choices. • It is fussy, and takes effort when you could be having fun coding or chatting to users. • In essence it is a representational device rather than a technique.	• Cognitive task analysis focuses on what the user knows and frequently is concerned with the quality of the mapping between a representation of the user's knowledge and that required by the system. • GOMS family	• Sometimes one wants to analyse what a system as a whole must do, not just what the user must do, or just what the system must do (though it is also called conceptual design). • This can be especially useful if one is designing a new way to do a familiar task, since the logical description should apply equally to the before and after systems.

Methods: GOMS

One of the first “Usability” methods (not so common today)

Quantification of the use efficiency of systems over time which require interaction. Foundations: systematically developed tables (“A mouse click takes X seconds, keyboard entries take N seconds”,...). Problem: **Decoupling of qualitative and quantitative aspects, separation of functionality and interface.**



Goals



Operations

Concrete operations (Keystrokes etc.)



Methods

Related sequence of operations



Selection Rules

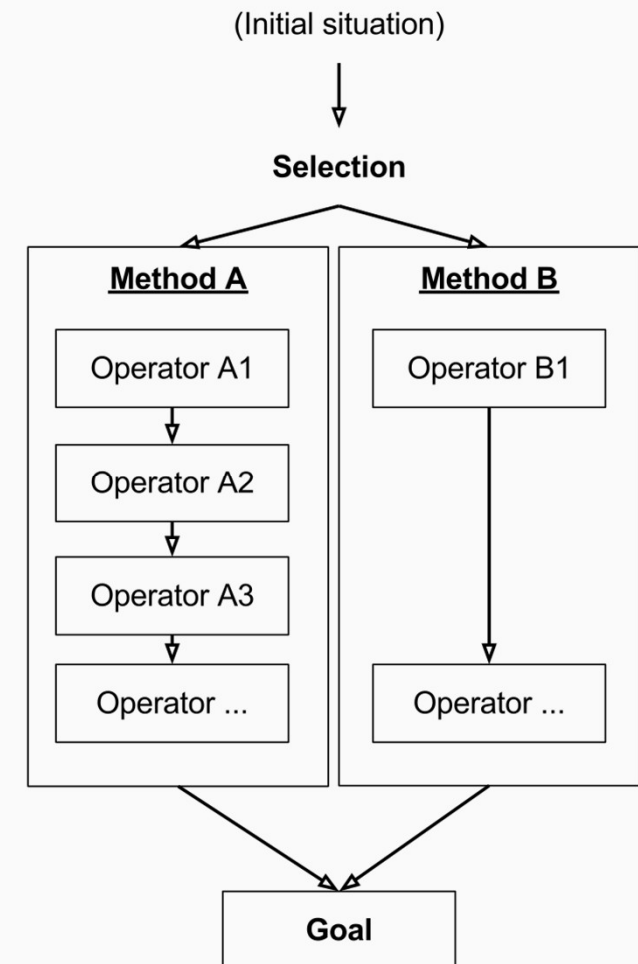
Choices

Selection

Method

Operator

Goal



Methods: GOMS

One of the first “Usability” methods (not so common today)

Quantification of the use efficiency of systems over time which require interaction. Foundations: systematically developed tables (“A mouse click takes X seconds, keyboard entries take N seconds”,...). Problem: **Decoupling of qualitative and quantitative aspects, separation of functionality and interface.**

- G** Goals
- O** Operations
Concrete operations (Keystrokes etc.)
- M** Methods
Related sequence of operations
- S** Selection Rules
Choices

(John & Kieras 1996)

Moving text with the MENU-METHOD

Description	Operator	Duration (sec)	
Mentally prepare by Heuristic Rule 0	M	1.35	mark text to be moved
Move cursor to beginning of phrase (no M by Heuristic Rule 1)	P	1.10	
Click mouse button (no M by Heuristic Rule 0)	K	0.20	
Move cursor to end of phrase (no M by Heuristic Rule 1)	P	1.10	
Shift-click mouse button (one average typing K)	K	0.28	TWO commands needed to complete a move. Should we consider a MOVE command instead?
(one mouse button click K)	K	0.20	
Mentally prepare by Heuristic Rule 0	M	1.35	issue CUT command
Move cursor to Edit menu (no M by Heuristic Rule 1)	P	1.10	
Press mouse button	K	0.10	indicate insertion point
Move cursor to Cut menu item (no M by Heuristic Rule 1)	P	1.10	
Release mouse button	K	0.10	issue PASTE command
Mentally prepare by Heuristic Rule 0	M	1.35	
Move cursor to insertion point	P	1.10	Issuing commands will be used a LOT! Can we shorten this procedure? Consider keyboard shortcuts.
Click mouse button	K	0.20	
Mentally prepare by Heuristic Rule 0	M	1.35	
Move cursor to Edit menu (no M by Heuristic Rule 1)	P	1.10	
Press mouse button	K	0.10	
Move cursor to Paste menu item (no M by Heuristic Rule 1)	P	1.10	
Release mouse button	K	0.10	
TOTAL PREDICTED TIME		14.38	

Fig. 2. A Keystroke-Level Model for moving the text in Figure 1. The notes on the right represent handwritten notes an analyst might add to the KLM to highlight ideas.

Methods: GOMS

Goal: Send an Email

Operators:

- K (Keystroke) for keyboard actions.
- P (Pointing) for mouse actions.
- H (Homework) for mental operations.

Methods:

M1: Compose Email

- K: Open email client (Ctrl + E)
- K: Create a new email (Ctrl + N)
- H: Think about the recipient's email address and type it.
- K: Compose the email content.
- K: Attach files (Ctrl + Shift + A if needed)
- K: Proofread the message.

M2: Send Email

- K: Review the email.
- P: Move the mouse pointer to the "Send" button.
- P: Click the "Send" button.
- H: Wait for a confirmation message.

M3: Manage Sent Items

- K: Save a copy in the Sent Items folder (Ctrl + S).
- K: Close the email client (Ctrl + W).

Selection Rules:

- Use the keyboard for text entry and shortcut commands.
- Use the mouse for pointing and clicking.
- Use mental operations for thinking and decision-making.

Sample Tasks

Online Shopping Checkout Process:

- Create a hierarchical task analysis for completing a purchase on an e-commerce website.
- Start from selecting an item in the shopping cart to completing payment.
- Consider subtasks like selecting a delivery address, choosing a payment method, and confirming the order.

Booking a Flight Online:

- Perform HTA on booking a flight through an airline website.
- Decompose tasks from searching for flights to completing payment.
- Identify conditional tasks, such as choosing one-way versus round-trip, selecting seats, and adding baggage.

Setting Up a New Email Account:

- Analyze the task of setting up a new email account on a web-based platform.
- Identify and break down steps, from filling in personal details to configuring security settings.
- Include optional tasks, such as enabling two-factor authentication.

Logging into a Website:

- Develop a GOMS model for logging into a website.
- Define the **goal** (accessing the account), **operators** (entering username, password, clicking “Log In”), and possible **methods** (using password autofill, manually entering credentials).
- Suggest **selection rules** based on conditions, such as using autofill when available or manually entering credentials if autofill is disabled.

Sending a Message on a Messaging App:

- Create a GOMS model for sending a message on a mobile messaging app.
- Define the **goal** (sending a message to a contact), **operators** (opening the app, selecting the contact, typing, pressing “Send”), and **methods** (typing versus voice input).
- Suggest **selection rules** based on factors like context (voice input if typing is not feasible, or typing if in a public place).

Saving a File in a Word Processor:

- Analyze the task of saving a document in a word processor using GOMS.
- Define the **goal** (save the document), **operators** (clicking “File,” choosing “Save As,” entering a filename), and **methods** (keyboard shortcuts or menu navigation).
- Identify **selection rules** based on user familiarity (experienced users might use shortcuts, while beginners use the menu).

The HCI Challenge

Basics

Analytic
Design/Implementation

Human computer interaction is a difficult endeavor with glorious rewards.

Designing interactive systems to be

effective
efficient
easy to use
enjoyable to use

*so that **people** and **society** may realize the benefits of computation-based devices*

How to go beyond platitudes like “know the user”?

Users and Use Context

Basics

Primary Users	Secondary Users	Tertiary Users
Those who directly interact with the technology	Those who use it via an intermediary; may have an influence about primary users (using/not using it)	Those who are affected by the technology use and decide its purchase/implementation

Use Context

- Where are users located?
- What are the social, cultural and organisational aspects of the context?
- With what/whom do they interact?
- What role do technologies play in the context?
- How do these aspects impact upon the way that users do things?

stakeholders

motivations, and goals

understand people's

behaviors

An individual or group with an interest in the success of a product/organization in delivering intended results and maintaining the viability of the organization's products and services. **Stakeholders influence** programs, products, and services (Kotonya and Sommerville, 1998)